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DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE

B.Sc. Computer Science & Information Systems

Projects Guidelines

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1. Introduction

This document outlines the guidelines to be followed by completing students as they undertake their final year development projects in Computer Science or in Information Systems, in the department of Mathematics and Computer Science (MACS).

The document presents the general report layout, which may be reviewed per the advice of individual supervisors. Different terminologies may be used to refer to similar content, such as Abstract and Summary. Efforts are therefore made to have these guidelines as an all-encompassing document, although students must still follow the variations to be recommended by their supervisors.

However, confusions must be avoided between terms that bear different contextual meanings. For instance, a summary and a conclusion are different and must be applied appropriately. The same can be said about a list of references versus a bibliography. They cannot be used interchangeably (Cooper & Schindler, 2006).

It is also important for students to observe both the content of each chapter and of those that are merged. The title of each project, and of each chapter, must portray the relevant content. It is the same case with the headings of tables, figures and diagrams (Mouton, 2001).

It is a responsibility of each project undertaker to know in advance the submission dates, as announced by the department. This also applies to the dates of consultations and submissions of updates to supervisors. It is therefore within the rights of supervisors to reject, defer or revise the project where the undertaker does not fulfil the expectations in undertaking the project. That is, either a student missed the deadlines or does not comply with the supervisor's recommendations.

The layout of the report is outlined in Section 2, which lists the key features that each report must include. These features are detailed in Section 3 to help the students' better understanding. Students must follow the stipulated *Report features* (Section 4) to ensure uniformity of all reports, and hence enable an objective assessment of the contents. The interactions between the project undertakers and the supervisors are highlighted in Section 5, albeit any variations will be provided by individual supervisors.

Each report must include the standard cover page (Appendix A). In addition, efforts are made to disambiguate some area of the report (Appendix B), although students are strongly encouraged to seek clarification earliest in their project undertaking.

2. Report layout

It is advisable to include any content that, without creating unnecessary confusion, may help assessors and the general readers to either understand the steps that led to a conclusion, or help convince them of the reliability of your method (Hofstee, 2006). The development project in particular will follow a range of lifecycle models, as dictated by the user requirements, resources such as time and tools, and the area of study (Sharp, Rogers & Preece, 2019). The following blueprint of all development projects should therefore be adapted for relevance in a specific field.

- Cover page
- Blank page
- Acknowledgements
- Blank page
- Summary / abstract
- Blank page
- Table of contents
- List of tables
- List of figures
- Report structure
  - Introduction
  - Literature review
  - Method (i.e. design, methodology)
  - Results
  - Conclusion
- List of references
- Appendices (diagrams, figures, data, code, etc.).

3. Details of the report layout

A. **Cover page:** This page is also known as the title page. It reflects your topic, which must contain keywords, be brief, and exclude any jargon. It must accurately reflect the theme of the project. Leave out unnecessary phrases such as "An investigation into…", "An exploration of…" Appendix A presents a template of a cover page, as approved by the department. This page should not be numbered, and must be followed by a blank page.

B. **Summary or abstract:** This may not exceed a page, if it has to go that far. It is normally written after all the report chapters are completed. The contents include (a) The problem that the project solves (b) A brief description of the methodology (c) A bit of the nature of the data and/or results from testing the project, and (d) an overview of the conclusions.

C. **Acknowledgements:** it is only honourable to acknowledge those who made meaningful contributions to your study. This does not alter your evident work. A limited degree of informality is acceptable, without going overboard. This is followed by a blank page.

D. **Table of contents, List of tables, List of figures, List of abbreviation,** etc.: Inserting a Table of contents is the last thing to do in a report. It is done after all the levels of headings to sections, tables, figure, diagrams, etc. are formatted and edited to satisfaction. The headings' words should not span over one line, where possible. Also, the levels of the headings must be limited to three, and with logical numbering. For instance, whereas bulleting is another form of numbering, they may not precede the numbers. Also, both numbers and bullets must be used consciously with regard to a need for ordering items, or lack thereof. Below is an example illustration:

Chapter 1: Introduction

1.1 Introduction
  1.1.1 Background / rationale
    A. Network coverage
    B. Skills of users
      (i) Reading skills
      (ii) Writing stills
        - Braille
        - Digital pen

1.2 Literature review

In addition, the headings of figures, diagrams, etc. must also appear in the Table of contents, after they are inserted. No manual entry may be done on the above lists. In a digital file, these must be clickable.

E. Report structure

i. **Introduction:** All good reports deserve a decent introduction chapter, one that *very* briefly presents the background of the study, the problem that the study intends to solve, how this will be done (methodology, including the lifecycle model to expect, etc.), the scope of the project, the time frame, and the general outline of the report. The author may not introduce any agenda (i.e. chapter) that is not mentioned by the introduction.

ii. **Literature review:** this presents the body of work as previously published by other scholars. It has to be comprehensive yet limited, critical and contextual. It should not, however, be a damping site where all the articles that the author has read are simply copied and pasted without sufficient nor critical review.

iii. **Method:** the contents will differ largely with the field of study, the nature of the project, and also be based on the solution to provide. Most development projects follow a standard lifecycle model, which must be outlined. The chapters to follow may also adapt the phases (or their combinations) of the chosen model.

iv. **Results:** the outcome of the undertaking must be presented in clear format or categories. The correctness, accuracy and authenticity of data must be preserved through the use of appropriate data collection techniques and results presentation techniques. E.g. A table lists intersecting categories of results. A graph does more than a mere presentation by visually showing the trends and comparisons that are easy to interpret. Well-presented results facilitate their easy discussion and conclusions.

v. **Discussions and conclusion:** in some cases, the discussions are done in the same chapter as the conclusions while, in other cases, the two are separate. This prompts a need to rename the chapters accordingly. Either way, the conclusions must be mapped to the objectives, the aim, and the problem statement of the study. For instance, if there are three (3) problems or objectives presented in Chapter 1, a conclusion to each must be discussed explicitly in this chapter.

   This chapter may also include a section on the future prospects, although some may as well rename the chapter in combination of the two.

   Hence, variations of the chapters range as follows:

- Result; Results discussions; Conclusions; Future prospects (for huge content)
- Results discussions; Conclusions and future prospects (for normal size).
- Results and discussions; Conclusions and future prospects (for normal size).

vi. **List of references:** referencing is done to acknowledge the content from other sources, that is, words, ideas, data or influence of other scholars on your work. Every entry in this list must have been included within the body of the report. There are a range of referencing styles, from Harvard referencing style, Chicago Manual of styles, Modern Language Association (MLA), to the widely used American Psychological Association (APA) style. Your department, or your supervisor at the least, will have prescribed a specific one. The listing of references, whether it is chronological, alphabetical or numbered by order of mention in the report, depends on the referencing style followed.

vii. **Appendices** (diagrams, figures, test results, code, etc.): All the important material that contributed to the completion of the project, but which does not feature in the flow of the report, can be categorised in this section. It is worth mentioning that expanding the size of the diagrams, tables or font size merely degrades the quality of a report. A long report does not justify its acceptability. In fact, it does the opposite!

4. Report features

i. **Document formatting:** The report formatting must conform to the requirements of the department, unless an individual supervisor expresses specific preferences that his/her students must follow. Such include:

Typeface (Times New Roman); body font size (12 pts); font colour (black); line spacing (1.5 pts); paragraph spacing (6 pts before and after); body alignment (Justified); table contents alignment (left align); list of references alignment (left, with hanging indent).

*** Do not use the Enter key to create any 'white space' between paragraphs. Use either the paragraph spacing or the page breaks, as it is necessary.*

ii. **Report length:** The length of a report will depend on the field of study, and the nature of the project. Nonetheless, students must bear in mind that the acceptability of the academic work depends solely on the quality of the content, and certainly NOT on the length or quantity of the content. As a matter of fact, a report with a lot of redundant and/or irrelevant content, enlarged diagrams, floating white space, and enlarged tables and figures does a lot more damage than good. Supervisors or any reader do not consider

academic reports as leisure novels, and hence prefer to invest their time only on worthwhile content.

As a result, keep the report body within 20 pages, and adjust the rest of the contents to meet the requirements of your supervisor. There must be a strong justification to submit a report that spans beyond 35 pages.

iii. **Submission mode**: All the work done in a development project must be documented and submitted to the department for assessment in electronic format (e.g. PDF, zipped files). That is, the periodic presentation slides, the final report, and any relevant electronic material (such as user and/or technical manuals) as it will be advised by the supervisors, must be submitted by e-mail to the respective supervisors.

5. Supervision

All supervising lecturers will attend the three presentation sessions and their assessments will contribute to the final marks.

A. Candidate projects

At the beginning of each academic year, the department will announce available supervisors in their its different fields of interest. Students are expected to identify their projects of interest (from a pool of project ideas as it will be provided by the department, or their own projects' ideas). The final decision must be approved by the department.

B. Consultation

Students are expected to schedule suitable consultation times with their respective supervisors, which is normally once every fortnight (two weeks) throughout the academic year. The respective supervisors may recommend a variety of communication modes for a specific phase of the project, which may range from physical to virtual meetings. Most supervisors keep journals of students' communication, which is an objective way to support decisions to be made at the end of an academic year.

C. Submission targets

Students are expected to make three projects presentations (Table 1) to the department during their final academic year (as a continuous assessment). All presentations must be approved by supervisors BEFORE submissions. The presentation sessions are done on Fridays. Specific dates will be announced by the projects' coordinator. Students are expected to dress in their best presentable ways, for a good academic impression.

Table 1: The contents of the three oral presentations

	First presentation: Project Proposal	**Second presentation:** Progress update	**Third presentation:** A complete project
1	On the second week of lectures in Semester A.	On the first week following the midterm break in Semester B	On the last day of lectures in Semester B
2	Project Title/Topic	Project Title/Topic	Project Title/Topic
3	Table of contents	Table of contents	Table of contents
4		Introduction	Introduction
5	Theoretical background		Theoretical background
6	Problem statement		Problem statement
7	Motivation		Motivation
8	Aim / objectives		Aim / objectives (* Match)
9	Scope and limitations		Scope and limitations
10	System requirements	Detailed system requirements	Detailed system requirements
11	High-level / conceptual system design (incl. lifecycle model)	Methodology (incl. lifecycle model)	Methodology (i.e. How?)
12	Tools and technologies to be used	Tools and technologies used	Tools & technologies used
13	Project schedule (start to finish dates)	Project schedule update (start to finish dates)	Project schedule update (start to finish dates)
14	Demonstration (of a low-fidelity prototype)	Demonstration (of a high-fidelity prototype)	Demonstration (of a final product)
15	Summary	Summary	
16			Conclusions (* Match)
17	Questions time (take notes)	Questions time (take notes)	Questions time (take notes)

D. Assessment criteria

The overall mark for the project will comprise the three presentations and a written report in the ratio of **30**:**70**. That is,

- Three oral presentations (proposal, progress, and final), each weighing 10 %.
- Written report (to be submitted two (2) days after the final project presentations), 70 %.

E. Project selection criteria

The supervisor(s) will prepare brief project descriptions and/or research interests. Students are also encouraged to submit their projects' ideas to the supervisor(s) for approval by the relevant supervisor(s).

F. Copyright

All submitted projects remain the intellectual property of The National University of Lesotho. The projects which attract the *NUL Innovation hub* will, under the assessment of the supervisors, be motivated for further development.

All students who undertake projects are bound by the contents of this Projects Guidelines, which will be reviewed before the start of each academic year, as it may be deemed necessary.

Appendix A: Cover page

The following page represents a standard cover page to be used by all projects students, with respective adjustments made where necessary.

<Title of a project>



A final year project submitted by

<Name(s) of student (s)> <Student number (s)> [, and]

In partial fulfilment to the requirements for

<B.Sc. Computer Science> | <B.Sc. Information Systems> |

To the

Department of Mathematics and Computer Science

Faculty of Science and Technology

National University of Lesotho.

<Month> <Year>

Supervisor: <Names>

Appendix B: Emphasis of some sections for clarity to students

Introduction

The structure of a typical project report is outlined below. It can and should vary to suit your specific project. You are not compelled to write text just to fill out each section described below. Add sections that are relevant to your project even if they are not included in this guide.

Note: This document is only a guide.

A report should be broken down into a series of chapters, with each chapter consisting of a sequence of appropriately numbered sections and sub-sections. A typical list of chapters for a design and programming scientific report is described here. (But remember, you can adapt or replace to fit the needs of your specific project)

List of items for a project report (in this order**)**

i. **Cover page: see appendix A.** A cover page should not be numbered!

ii. **Acknowledgements**

Pass a vote of thanks to everyone (individuals or groups) who provided advice and support to your project. Use Roman numbering for this page – likely to be Page (i).

iii. **Abstract (**half a page or 300 words long**)**

An abstract is a concise summary of the project/research that describes the content and scope of the work. It identifies the main objectives, the methodology, key findings/results, and the conclusions. It is meant to summarize the major aspects of the entire project report in a prescribed sequence that generally includes: (a) the overall purpose of the project and the problem statement under investigation. (b) the design of the study. (c) major findings or trends within the study, and (d) a brief outline of the results interpretations and conclusions.

Advice: An abstract should be the last thing to write because it is meant to summarize the contents of the report.

Use Roman numbering for this page – likely to be page (ii)

iv. **Table of contents**

Provide a Table of contents with no more than three levels, and with page numbers. Use Roman numbering for this page – likely to start from page (iii). It may, however, extend depending on the number of pages that the Table of contents covers.

v. **List of figures and tables**

If the report has many figures and tables, their list is provided here, and it includes their page numbers. Use Roman numbering for this page – it should appear immediately under the last item in the table of contents.

vi. **Introduction (**2 – 3 pages**)**

Outline the problem in hand, why it is interesting/important and what the challenges are.

List the aims/objectives and goals.

- An *aim/objective* is something intended to be achieved (e.g. Learning a new programming language or software engineering concept and then applying it in solving a problem).
- A *goal* is something specific that is expected to be delivered (e.g. a working application with a particular set of features).
- An overview of how the project will be carried out (e.g. following an Agile software process / a formal software process, iterative approach etc.).
- A brief overview of the rest of the chapter in the report (a guide to a reader of the overall structure of the rest of the report).

This chapter is relatively short and must leave the reader very clear on what the project is all about and what the goals and objectives are. The numbering should start on this page.

vii. **Literature review / study (**2 - 4 pages**)**

- This chapter should cover the related work, research done by other scholars, and an overview of the tools to be use in the project.
- If relevant, provide a survey of similar solutions, programs or applications/systems, and how the current project will differ from the existing solutions.
- A description of the research carried out to learn about the nature of the problems under investigation and the potential solutions. The methodology will vary with projects. For example, it might involve searching through research publications and online resources, or it might involve an exploration of the design possibilities for a user interface (UI) or program structure.
- Outline and reference the sources of information drawn from, (papers, books, websites, etc.), stating how each relates to the work.
- Introduce the software, programming languages, library code, APIs and frameworks, and other tools to be used. Discuss choices and clarify the ones to be used.

  Note: DO NOT include well known concepts (e.g. HTML, Java, C# etc.) or try to give tutorials on how to use a tool or code library (use references to books and websites for that kind

of information). Everything included should be directly relevant to this work and the relationship be clear.

viii. **Requirements / specification (**1 – 2 pages**)**

- Give detailed problem statement. This must elaborate what you may have included in the Introduction chapter, and represents the starting point from which requirements were derived.
- Provide a list of requirements.
- Apply the Use-case diagrams to present requirements. Provide detailed Use cases and Use-case diagrams in the Appendices.
- Provide the results of analysing the requirements to extract information. E.g. data modelling to find the data to be stored (ER diagram), views/ web pages needed and so on.

  Note: The level of details of the requirements and Use cases will depend on the nature of the project.
- For a Software-engineering based design and implementation project, the requirements must be thoroughly defined.

ix. **Design and implementation** (4 – 6 pages)

- Describe the design of the project.
- Start with the application architecture, the overall structure and the relevant components.
- Describe of the design of each component of the architecture.
- Include the database or storage representation.
- Provide the implementation details as necessary.

  Note: Like other chapters, a structure and the contents of this chapter will depend on the nature of the project (the above list is a suggestion and not a fixed requirement).
- Use the appropriate terminology to describe the design – focusing on the most interesting design decisions. For example, what were the alternatives? Why was a particular solution selected over others?

  Note: The limited number of pages should be considered in deciding on the degree of the details. The use of appropriate terminology, and evidence of a good understanding of Computer Science/Engineering principles are imperative.
- The diagrams and screen shots are important to explain the design, albeit sparingly. Also, they should add information rather than duplicate what is in the text. The featuring of diagrams on consecutive pages must be avoided to preserve a logical flow of the report. Where

necessary, UML diagrams can be used, albeit sparingly. Else, any additional diagrams can be included in the appendix section.

- It is useful to include the code snippets to highlight how a particular algorithm was implemented or how an *interesting* programming problem was solved. However, lengthy sections of code must be avoided. Also, the code fragments must be readable, easy to follow and be properly indented. It may be better to feature a pseudo-code rather than the actual code, especially when describing an algorithm. Any long lines of code can be featured in the appendix and be cross-referenced.

x. **System testing** (2 – 4 pages)

- Describe the test strategy, that is, the unit, functional, acceptance testing, how they were carried out, and how the tests were selected?
- Include the examples of specific tests and how they were carried out. Focus on interesting cases/parts of the project.
- Include a summary of the test results and what coverage was achieved. The detailed tests' report(s) should appear in the appendices.
- If the project requires substantial evaluation of data and results, or other forms of testing that are not code/program-based, then this chapter must be adapted to the project.

Results and discussion (1 – 3 pages)

- Present all results obtained, discuss and evaluate them.
- Discussion the means of interpretations (what the results mean) the result obtained.

xi. **Future work** (1 page)

- How could the project be developed, given additional 6 months or so?
- What additional features or functionalities would be recommended to make the system more complete, useful, usable and more secure?

xii. **Conclusion** (1 – 2 pages)

A conclusion presents what the project has achieved. A *goal/objective-based* approach is recommended for this, to address each goal and objective set out in the introduction chapter. These should be individually evaluated and concluded on. The conclusion must also include a critical evaluation of the results of the project. e.g.

- Elaboration of how well were the goals/objectives met, and the application's *fit for purpose*.
- Whether a good design and implementation practice has been followed.

- A discussion of whether the right implementation technology, the right frameworks and APIs were chosen.
- A paragraph that wraps up and presents the final thoughts of the project.

xiii. **References** (at least 1 page**)**

Give a numbered list of publication details for all the items referenced in the report.

xiv. **Appendix**

Add the appendices section that are relevant to the project. A source code listing should always be included for a computing project, or an equivalent listing if a project has other kinds of outputs.

Note:

- The appendices should provide additional relevant information to the readers. This should not necessarily be part of the main chapters, and hence be in the appendices.
- Be selective on inclusions and avoid dumping unimportant or trivial information.
- Properly number your list of appendices.

The following are also important appendices for a computing project:

xv. **System Manual**

This should include all technical details that would enable another student/person to pursue this project in future, and be able to amend or extend the code. For example, the location of the code, and how it should be compiled and installed?

xvi. **User Manual**

This should give enough information for someone to use what is designed and implemented. It is a good place to include the screen shots of the application or the screen dumps.

xvii. **Supporting documentation or data**

Any additional data, diagrams or Use-case descriptions that are relevant to the documentation of this work can be included in the appendix section as supporting documentation or data.

xviii. **Test results and test reports**

The test results that add to the value of the report, but which would not fit within the page limit of the main report, can be included as the appendices.

xix. **Source code listing**

The source code should be properly presented and neatly formatted. Apply coding standards, comment on the code, without long lines of code arbitrarily wrapping round to the next line, that is, avoid a messy code. The page orientation can always be toggled between Portrait and Landscape to maximise the use of space. In addition, the number of columns can also be extended.

Appendix C: Outcomes and assessment criteria

Table 2: The project outcomes and assessment criteria

OUTCOMES	ASSESSMENT CRITERIA
a) Draft a project/research proposal showing a need for a software artefact in a specific stream (i.e. CS, IS).	▪A detailed proposal illustrating the research project idea is drafted. ▪There is a clear understanding of the research project and a problem to solve.
b) Produce a revised and detailed project proposal after evaluation of the draft proposal by the supervisor(s)	▪Problem background and statement are successfully documented. ▪A System Requirements Specification (SRS) is outlined ▪Requirements Gathering is properly done with prospective/intended end-users ▪Functional requirements are outlined ▪Use case scenarios and diagrams are effectively completed ▪Realistic Project Management timeline (Gantt Chart) is drawn ▪Students accurately identify methods, algorithms, tools etc., to utilize in the design and implementation phases ▪A formal proposal document is submitted to the respective supervisor and a copy to the project's coordinator before the set deadline.
c) Produce a preliminary design document for the project and evaluate the existing literature, systems/solutions that are similar to the proposed/developed	▪Preliminary Class diagrams are designed using UML from the use case diagrams ▪Preliminary test plans based on the Use-case scenarios are composed. ▪An outline of existing literature and systems, their strong and weak points are outlined (Literature review). ▪A formal deliverable document is submitted to the respective supervisor and a copy to the projects coordinator before deadline.
d) Produce a refined, detailed design document	▪A refined and detailed design document is produced ▪Interaction sequence diagrams are drawn using correct notation and design principles (i.e. Object Oriented Design) ▪Refined Class diagrams are produced using correct notation and design principles (i.e. Object Oriented Design) ▪A formal deliverable document is submitted to the respective supervisor and a copy to the project's coordinator before deadline.
e) Produce a prototype system	▪A prototype system is successfully demonstrated to the supervisor(s) ▪25% of the objectives are met
f) Provide an Analysis of Algorithms (applicable to CS specific projects) ▪ Provide proof of information retrieval (IR), data manipulation techniques, data and database management, using latest techniques ▪ Provide proof of networking concepts, algorithms, engineering and/or scientific methods, simulations or practical implementations. ▪ Document comments obtained during the prototype execution	▪ Detailed analysis of the algorithms being used in the project is outlined. ▪A comparison of the existing algorithms with applied ones is satisfactorily documented. ▪Proof of information retrieval (IR) from an information system – depicts input through a GUI, (processing) and output ▪Latest networking concepts in a field of study are demonstrated. Use of algorithms for Networking is successful. Engineering and/or Scientific methods are adequately shown ▪A simulation of the implementation is done and/or a practical solution exists ▪A formal document with screen-shots of the system executing or a simulation is submitted to the supervisor(s) and a copy to the projects coordinator.
g) Develop an Alpha version of the system and demonstrate to supervisor(s).	▪An Alpha version of the system is produced, demonstrated to the supervisor(s) ▪The Alpha version meets 50% of the intended objectives
h) Develop a Beta version of the system and demonstrate to supervisor(s).	▪A Beta version of the system is produced, demonstrated to the supervisor(s) ▪The Beta version meets 75% of the intended objectives
i) Demonstrate a fully functional system and submit the final documentation.	▪A fully operational system is produced and demonstrated. ▪Project documentation based on the MACS guidelines is produced and e-mailed. ▪An oral, power-point presentation is conducted in front of a panel of assessors. ▪The electronic package of the report with the source code, appendices, databases and presentations is submitted to the supervisor(s) and projects coordinator.

References

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